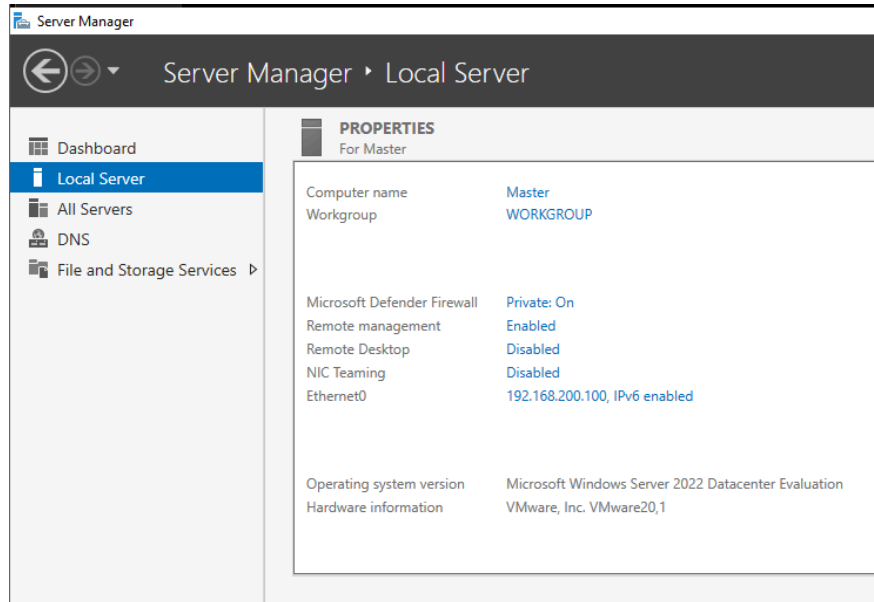


Windows Lab 3

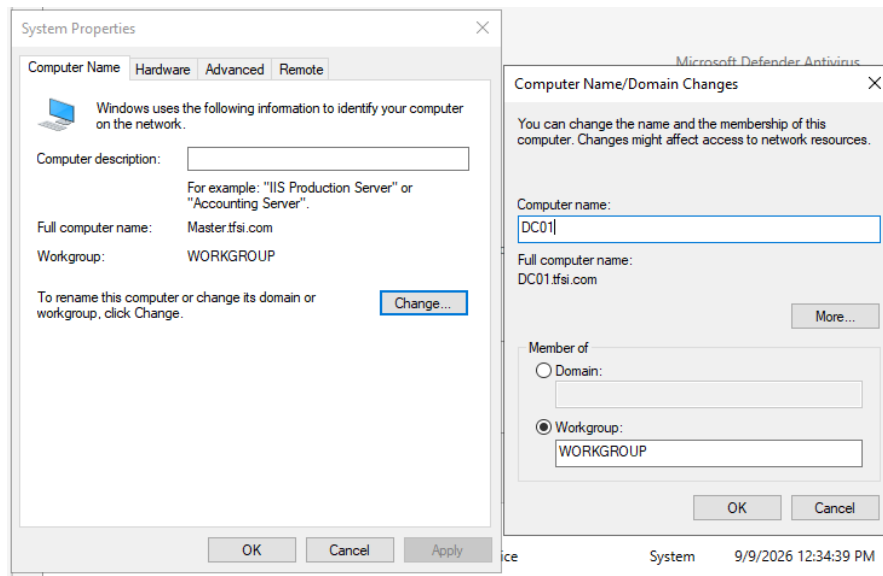
By Jai Pandey (260840140011)

Question 1

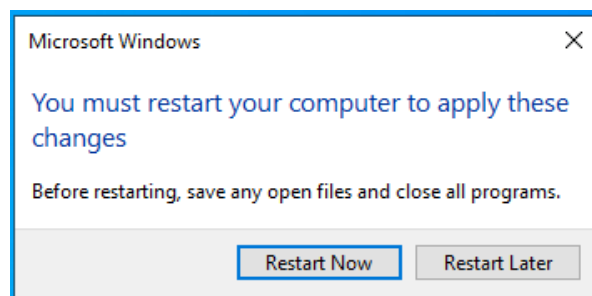
To change the computer name, Server Manager > Local Server > Computer Name



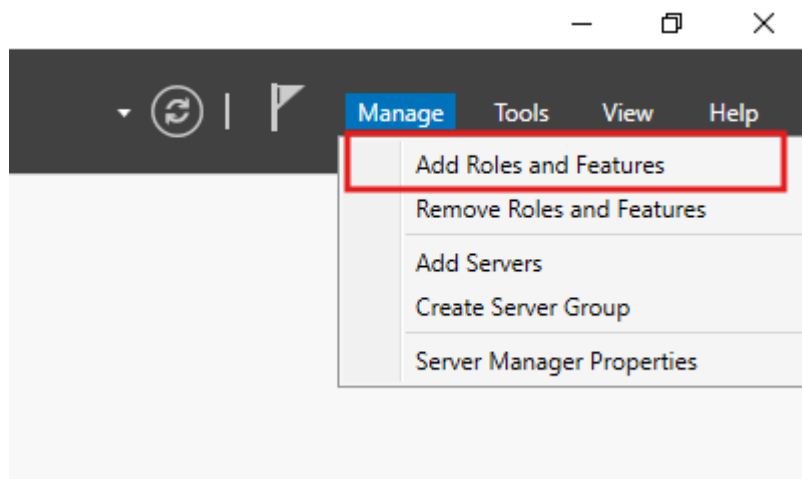
Click on "Change" Then enter the computer name, and click on OK



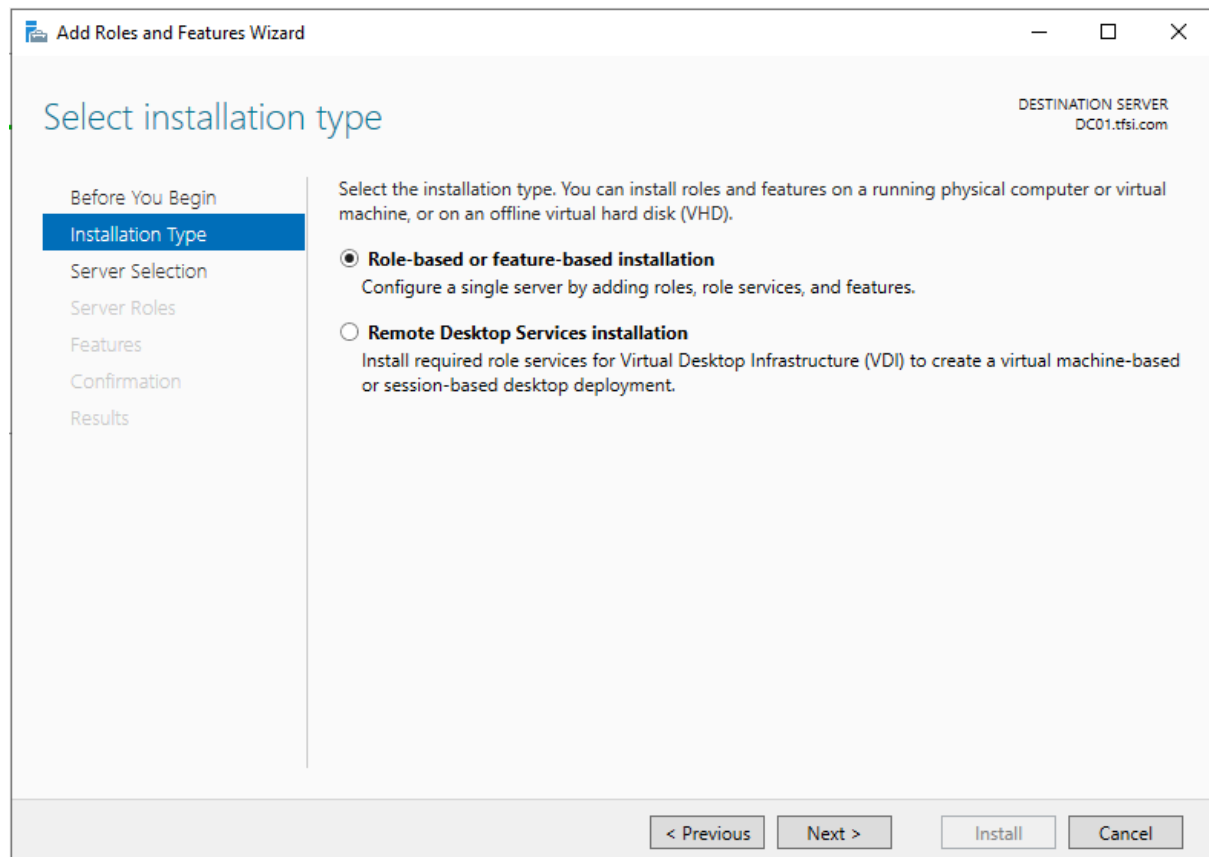
Now restart your server.



To make this the Active Directory Domain server
Go to Manage > Add Roles & Features



Select Role-based or Feature based installation



Select this server as the destination server

The screenshot shows the 'Add Roles and Features Wizard' window. The title bar says 'Add Roles and Features Wizard'. The main heading is 'Select destination server'. In the top right corner, it says 'DESTINATION SERVER DC01.tfsi.com'. On the left, there is a navigation pane with the following items: 'Before You Begin', 'Installation Type', 'Server Selection' (highlighted in blue), 'Server Roles', 'Features', 'Confirmation', and 'Results'. The main area contains the text 'Select a server or a virtual hard disk on which to install roles and features.' Below this are two radio buttons: 'Select a server from the server pool' (selected) and 'Select a virtual hard disk'. Under the selected option is a 'Server Pool' section. It has a 'Filter:' text box. Below the filter is a table with three columns: 'Name', 'IP Address', and 'Operating System'. The table contains one row: 'DC01.tfsi.com', '192.168.200.100', and 'Microsoft Windows Server 2022 Datacenter Evaluation'. Below the table, it says '1 Computer(s) found'. A paragraph follows: 'This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.' At the bottom, there are four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

DESTINATION SERVER
DC01.tfsi.com

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results

Select a server or a virtual hard disk on which to install roles and features.

☒ Select a server from the server pool
☐ Select a virtual hard disk

Server Pool

Filter:

Name	IP Address	Operating System
DC01.tfsi.com	192.168.200.100	Microsoft Windows Server 2022 Datacenter Evaluation

1 Computer(s) found

This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.

< Previous Next > Install Cancel

Select the Active Directory Domain Server option

The screenshot shows the 'Add Roles and Features Wizard' window. The title bar says 'Add Roles and Features Wizard'. The main heading is 'Select server roles'. In the top right corner, it says 'DESTINATION SERVER DC01.tfsi.com'. On the left, there is a navigation pane with the following items: 'Before You Begin', 'Installation Type', 'Server Selection', 'Server Roles' (highlighted in blue), 'Features', 'Confirmation', and 'Results'. The main area contains the text 'Select one or more roles to install on the selected server.' Below this is a 'Roles' section with a list of roles. The roles are: 'Active Directory Certificate Services', 'Active Directory Domain Services' (highlighted in blue), 'Active Directory Federation Services', 'Active Directory Lightweight Directory Services', 'Active Directory Rights Management Services', 'Device Health Attestation', 'DHCP Server', 'DNS Server (Installed)' (checked), 'Fax Server', 'File and Storage Services (1 of 12 installed)' (expanded), 'Host Guardian Service', 'Hyper-V', 'Network Controller', 'Network Policy and Access Services', 'Print and Document Services', 'Remote Access', 'Remote Desktop Services', 'Volume Activation Services', 'Web Server (IIS)', and 'Windows Deployment Services'. To the right of the roles list is a 'Description' section. It contains the text: 'Active Directory Domain Services (AD DS) stores information about objects on the network and makes this information available to users and network administrators. AD DS uses domain controllers to give network users access to permitted resources anywhere on the network through a single logon process.' At the bottom, there are four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

DESTINATION SERVER
DC01.tfsi.com

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results

Select one or more roles to install on the selected server.

Roles

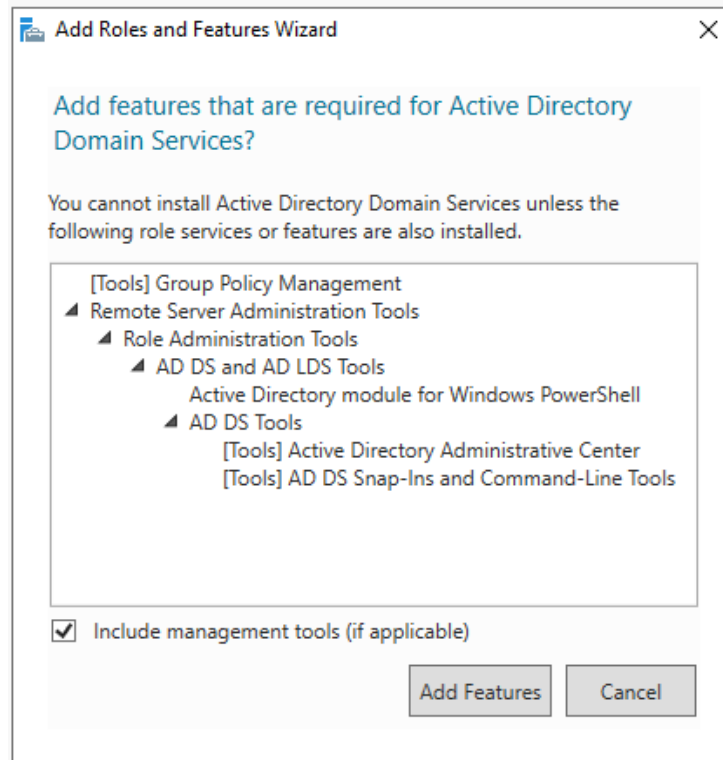
- ☐ Active Directory Certificate Services
- ☒ **Active Directory Domain Services**
- ☐ Active Directory Federation Services
- ☐ Active Directory Lightweight Directory Services
- ☐ Active Directory Rights Management Services
- ☐ Device Health Attestation
- ☐ DHCP Server
- ☒ DNS Server (Installed)
- ☐ Fax Server
- ☒ File and Storage Services (1 of 12 installed)
 - ☐ Host Guardian Service
 - ☐ Hyper-V
 - ☐ Network Controller
 - ☐ Network Policy and Access Services
 - ☐ Print and Document Services
 - ☐ Remote Access
 - ☐ Remote Desktop Services
 - ☐ Volume Activation Services
 - ☐ Web Server (IIS)
 - ☐ Windows Deployment Services

Description

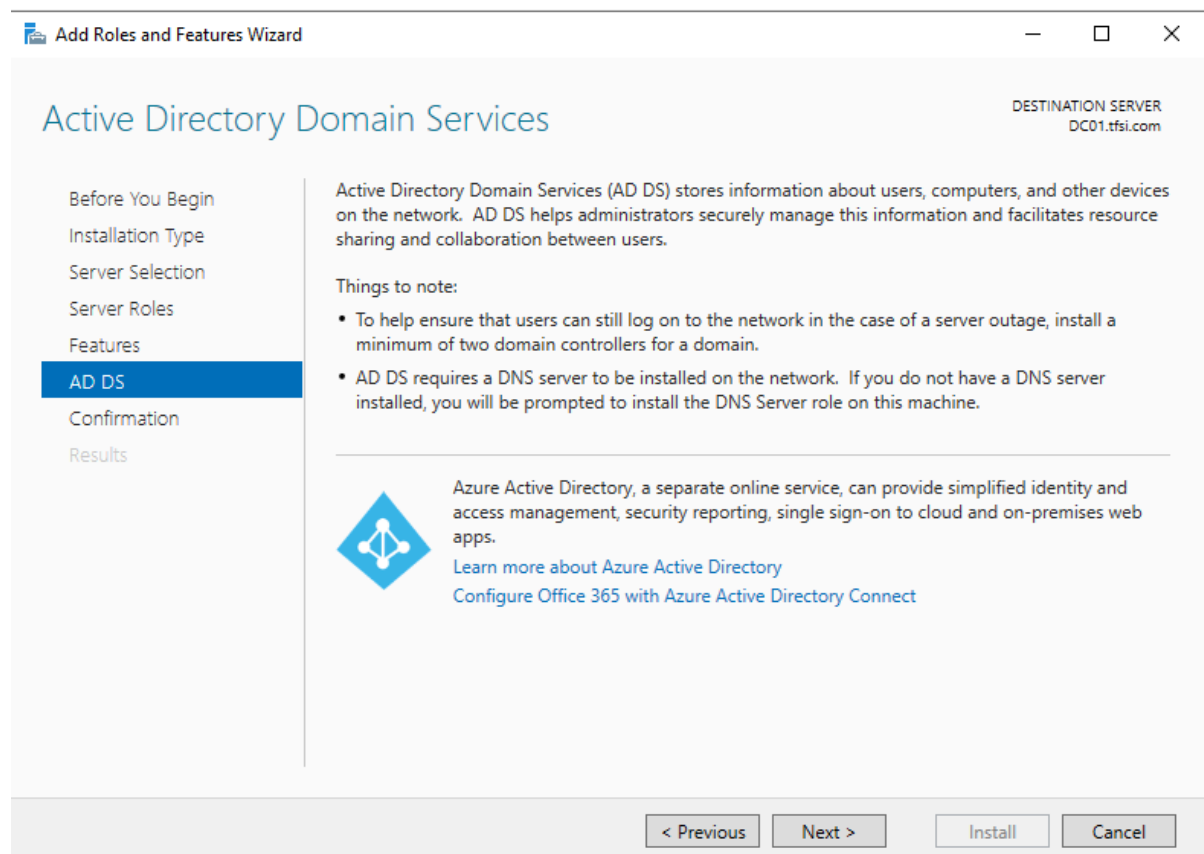
Active Directory Domain Services (AD DS) stores information about objects on the network and makes this information available to users and network administrators. AD DS uses domain controllers to give network users access to permitted resources anywhere on the network through a single logon process.

< Previous Next > Install Cancel

Click on "Add Feature" in the pop-up.



Click on next to in the info screen.



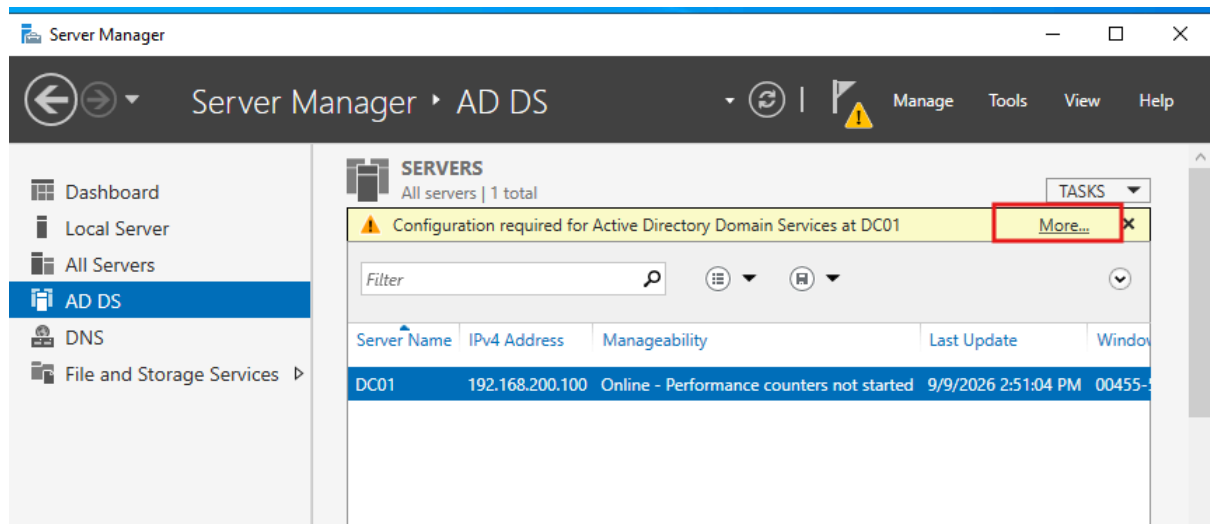
Preview selection, Select the Restart the destination server and proceed to installation.

The screenshot shows the 'Add Roles and Features Wizard' window. The title bar reads 'Add Roles and Features Wizard'. The main heading is 'Confirm installation selections'. In the top right corner, it says 'DESTINATION SERVER DC01.tfsi.com'. On the left, there is a navigation pane with the following items: 'Before You Begin', 'Installation Type', 'Server Selection', 'Server Roles', 'Features', 'AD DS', 'Confirmation' (which is highlighted in blue), and 'Results'. The main content area has a sub-heading 'To install the following roles, role services, or features on selected server, click Install.' Below this, there is a checkbox labeled 'Restart the destination server automatically if required' which is checked. A paragraph follows: 'Optional features (such as administration tools) might be displayed on this page because they have been selected automatically. If you do not want to install these optional features, click Previous to clear their check boxes.' Below this paragraph is a list of features: 'Active Directory Domain Services', 'Group Policy Management', 'Remote Server Administration Tools', 'Role Administration Tools', 'AD DS and AD LDS Tools', 'Active Directory module for Windows PowerShell', 'AD DS Tools', 'Active Directory Administrative Center', and 'AD DS Snap-Ins and Command-Line Tools'. At the bottom of the main content area, there are two links: 'Export configuration settings' and 'Specify an alternate source path'. At the bottom of the window, there are four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

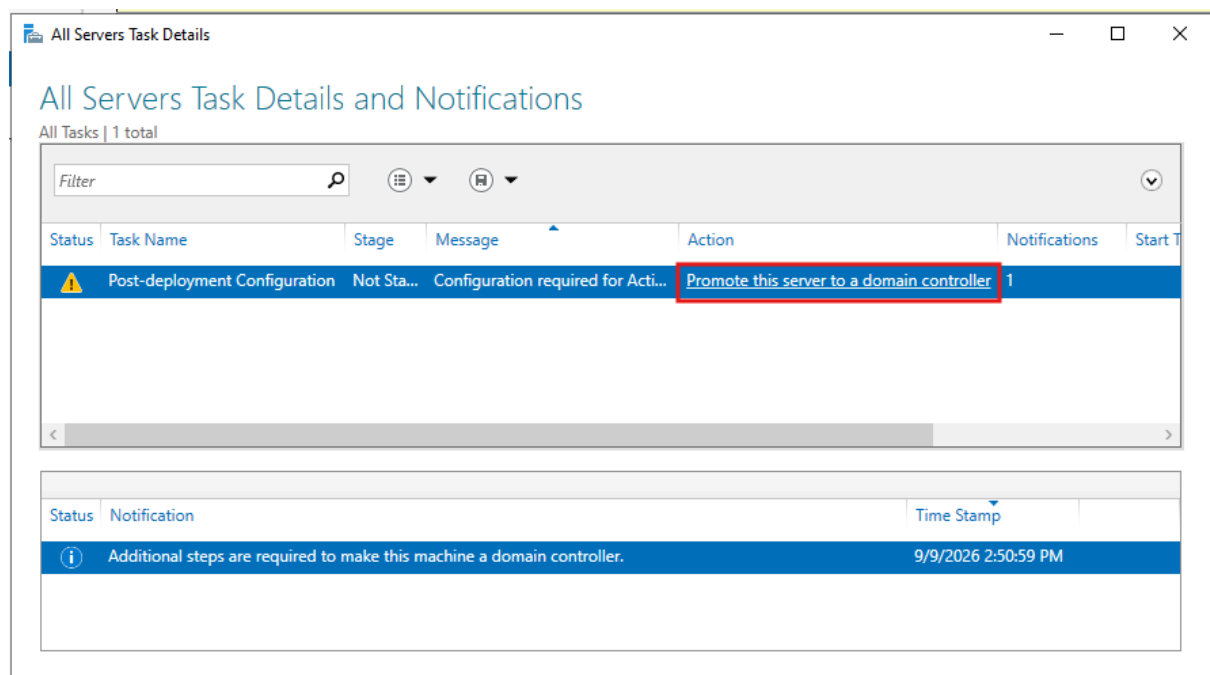
Restart after the installation in finished.

The screenshot shows the 'Add Roles and Features Wizard' window. The title bar reads 'Add Roles and Features Wizard'. The main heading is 'Installation progress'. In the top right corner, it says 'DESTINATION SERVER DC01.tfsi.com'. On the left, there is a navigation pane with the following items: 'Before You Begin', 'Installation Type', 'Server Selection', 'Server Roles', 'Features', 'AD DS', 'Confirmation', and 'Results' (which is highlighted in blue). The main content area has a sub-heading 'View installation progress'. Below this, there is an information icon (i) and the text 'Feature installation'. A progress bar is shown, which is mostly filled with blue. Below the progress bar, it says 'Configuration required. Installation succeeded on DC01.tfsi.com.' Below this is a list of features: 'Active Directory Domain Services', 'Group Policy Management', 'Remote Server Administration Tools', 'Role Administration Tools', 'AD DS and AD LDS Tools', 'Active Directory module for Windows PowerShell', 'AD DS Tools', 'Active Directory Administrative Center', and 'AD DS Snap-Ins and Command-Line Tools'. Below the list, there is a message box with a question mark icon and the text: 'You can close this wizard without interrupting running tasks. View task progress or open this page again by clicking Notifications in the command bar, and then Task Details.' Below the message box, there is a link: 'Export configuration settings'. At the bottom of the window, there are four buttons: '< Previous', 'Next >', 'Close', and 'Cancel'.

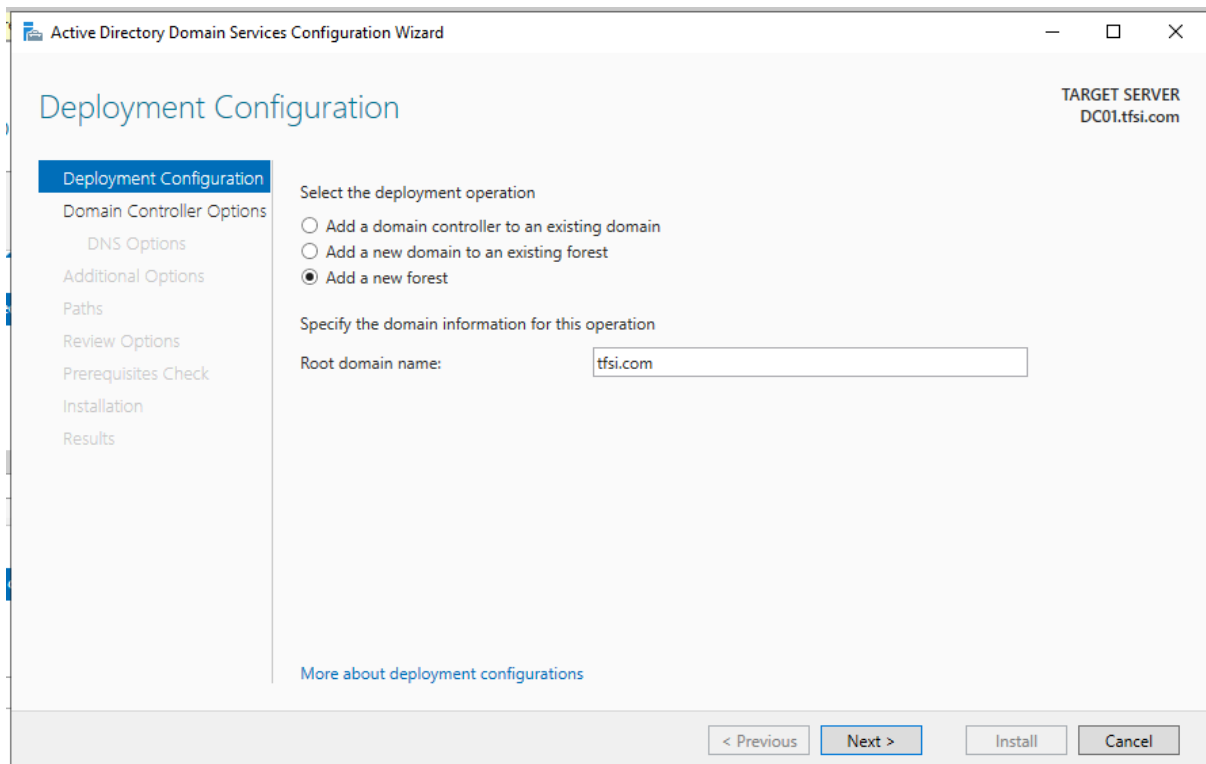
To promote the server as AD Domain Controller, Click on “More” in the yellow banner in the Server Manager > AD DS



Now select the “promote this server” option in the Actions.



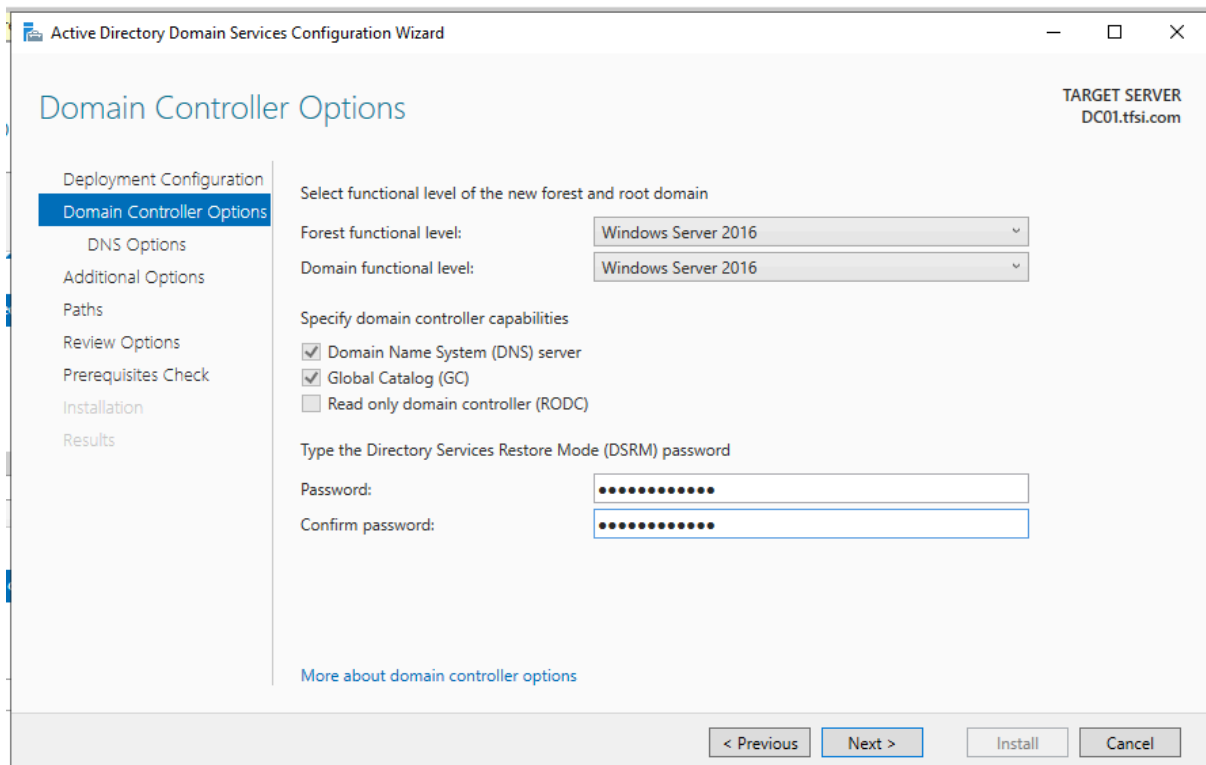
In the Deployment Configuration, Select the “Add a new Forest” option and enter the Root Domain name.



The screenshot shows the 'Deployment Configuration' step of the Active Directory Domain Services Configuration Wizard. The left sidebar lists the steps: Deployment Configuration (selected), Domain Controller Options, DNS Options, Additional Options, Paths, Review Options, Prerequisites Check, Installation, and Results. The main area is titled 'Deployment Configuration' and contains the following elements:

- Target Server:** DC01.tfsi.com
- Select the deployment operation:** Three radio buttons are present: 'Add a domain controller to an existing domain', 'Add a new domain to an existing forest', and 'Add a new forest' (which is selected).
- Specify the domain information for this operation:** A text box labeled 'Root domain name:' contains the value 'tfsi.com'.
- Buttons:** '< Previous', 'Next >', 'Install', and 'Cancel'.

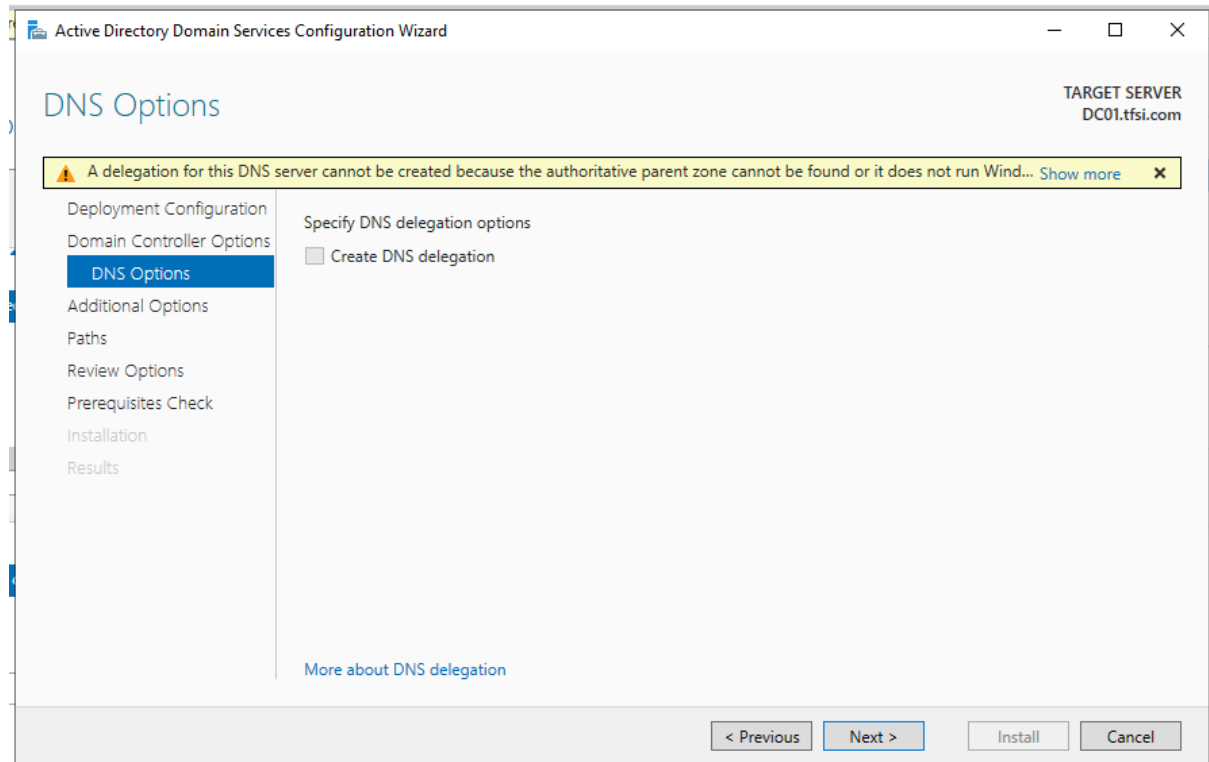
Enter the info as show in the form and click on next.



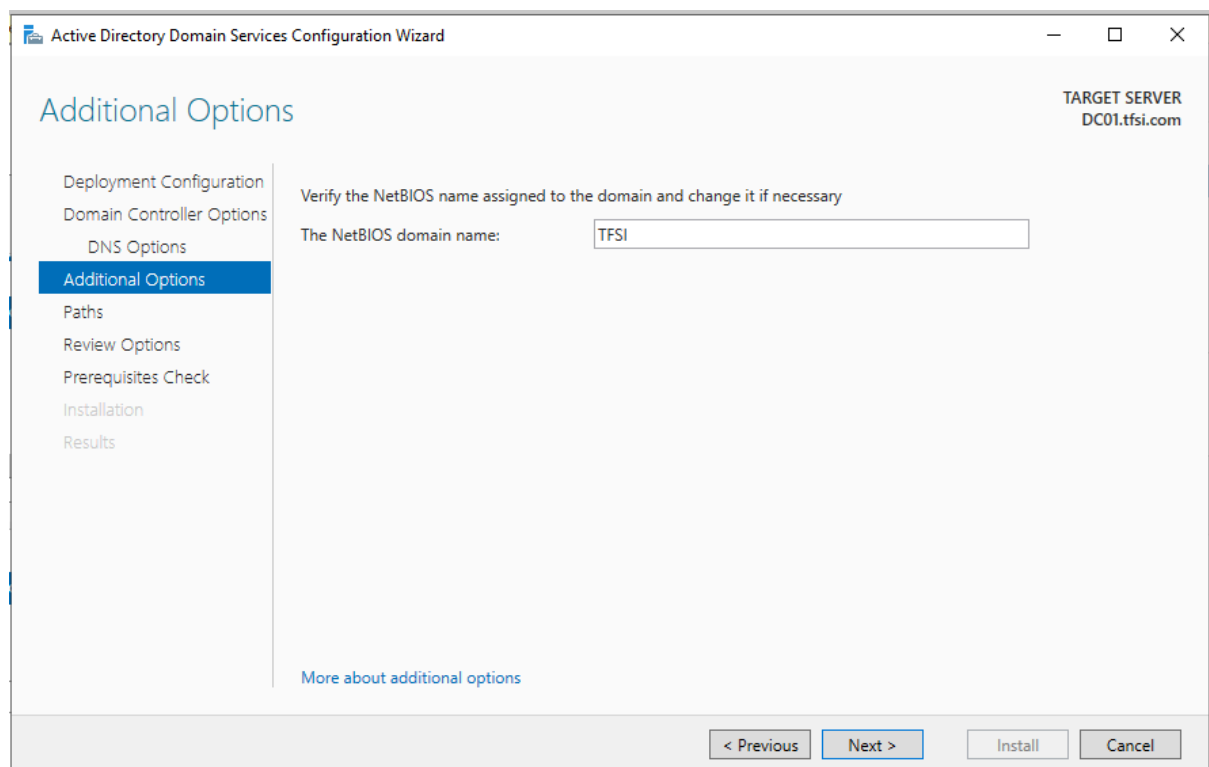
The screenshot shows the 'Domain Controller Options' step of the Active Directory Domain Services Configuration Wizard. The left sidebar lists the steps: Deployment Configuration, Domain Controller Options (selected), DNS Options, Additional Options, Paths, Review Options, Prerequisites Check, Installation, and Results. The main area is titled 'Domain Controller Options' and contains the following elements:

- Target Server:** DC01.tfsi.com
- Select functional level of the new forest and root domain:** Two dropdown menus are present: 'Forest functional level:' and 'Domain functional level:', both set to 'Windows Server 2016'.
- Specify domain controller capabilities:** Three checkboxes are present: 'Domain Name System (DNS) server' (checked), 'Global Catalog (GC)' (checked), and 'Read only domain controller (RODC)' (unchecked).
- Type the Directory Services Restore Mode (DSRM) password:** Two text boxes labeled 'Password:' and 'Confirm password:' are present, both containing masked characters (dots).
- Buttons:** '< Previous', 'Next >', 'Install', and 'Cancel'.

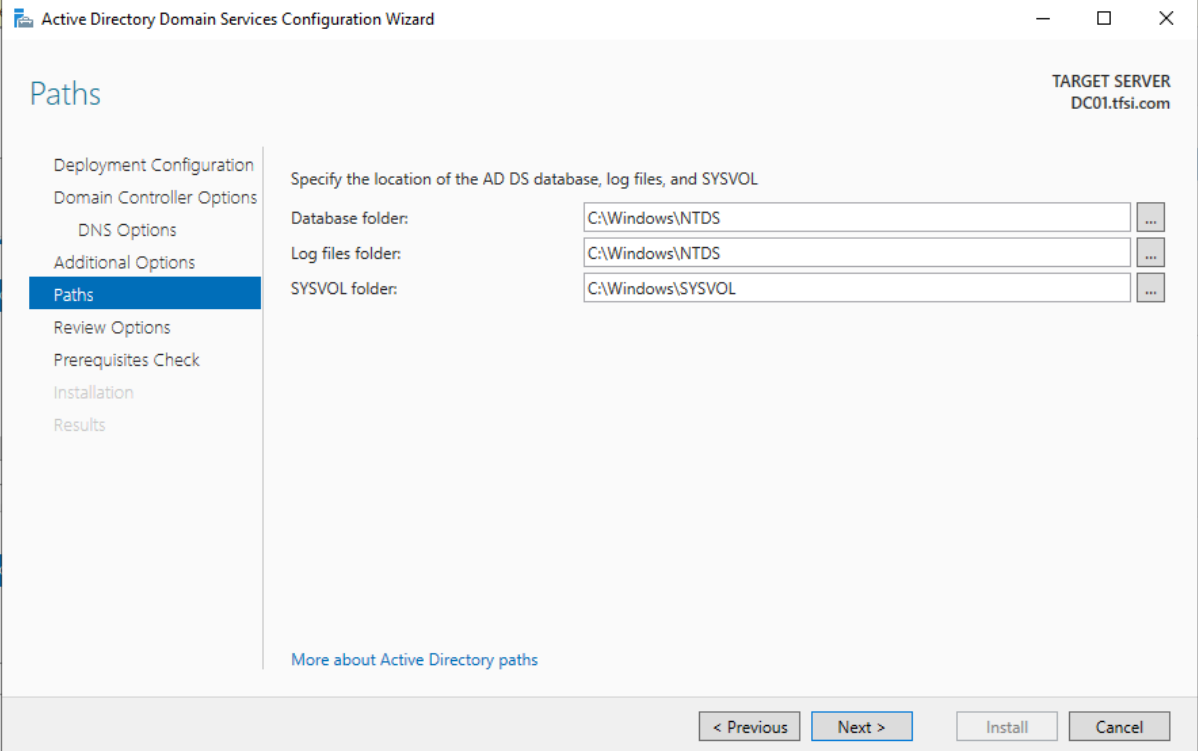
As this server is out DNS server as well, dont select anything in the DNS delegation step.



Enter the NetBIOS domain name.



Click on next to select the default paths for Database, Logs & SYSVOL folders.



The screenshot shows the 'Paths' step of the Active Directory Domain Services Configuration Wizard. The left sidebar contains a list of steps: Deployment Configuration, Domain Controller Options, DNS Options, Additional Options, Paths (highlighted), Review Options, Prerequisites Check, Installation, and Results. The main area is titled 'Paths' and contains the instruction 'Specify the location of the AD DS database, log files, and SYSVOL'. Below this, there are three input fields: 'Database folder:' with the value 'C:\Windows\NTDS', 'Log files folder:' with the value 'C:\Windows\NTDS', and 'SYSVOL folder:' with the value 'C:\Windows\SYSVOL'. Each field has a browse button (three dots) to its right. At the bottom right, there are buttons for '< Previous', 'Next >', 'Install', and 'Cancel'. The target server is identified as 'DC01.tfsi.com'.

Active Directory Domain Services Configuration Wizard

TARGET SERVER
DC01.tfsi.com

Paths

Specify the location of the AD DS database, log files, and SYSVOL

Database folder: C:\Windows\NTDS

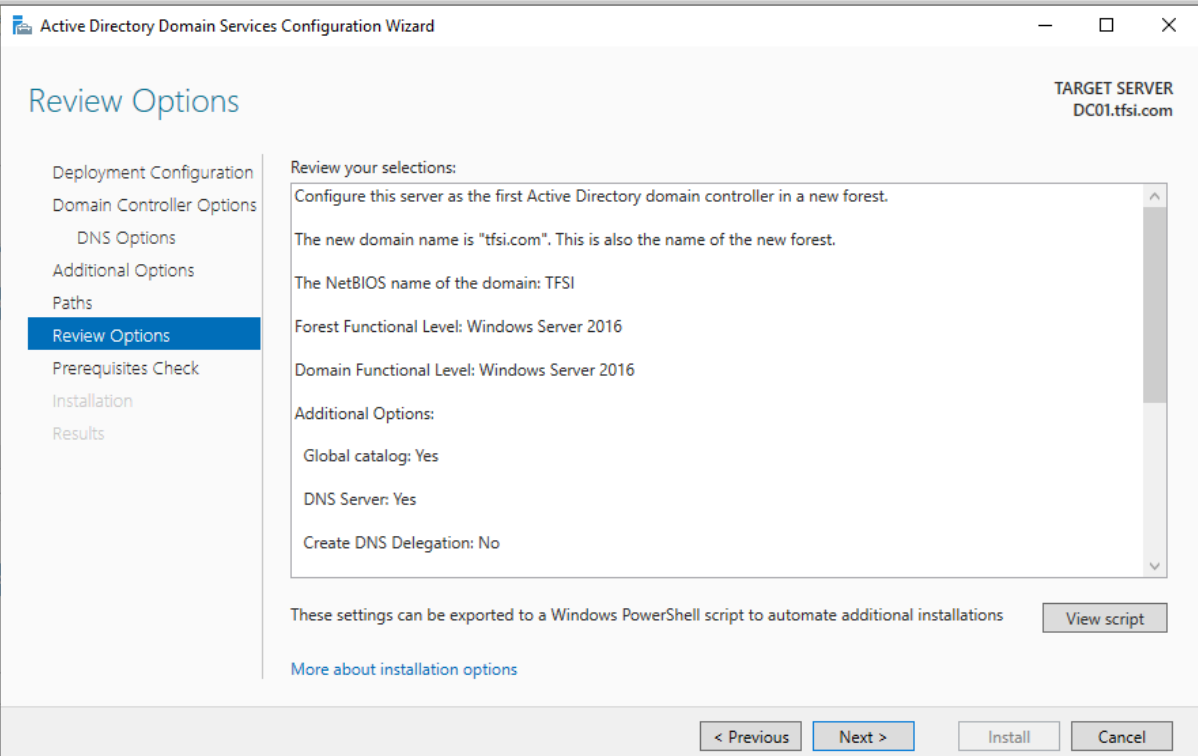
Log files folder: C:\Windows\NTDS

SYSVOL folder: C:\Windows\SYSVOL

[More about Active Directory paths](#)

< Previous Next > Install Cancel

Preview and click on next.



The screenshot shows the 'Review Options' step of the Active Directory Domain Services Configuration Wizard. The left sidebar is the same as the previous screen, with 'Review Options' highlighted. The main area is titled 'Review Options' and contains the instruction 'Review your selections:'. Below this, there is a large text box containing the following information: 'Configure this server as the first Active Directory domain controller in a new forest.', 'The new domain name is "tfsi.com". This is also the name of the new forest.', 'The NetBIOS name of the domain: TFSI', 'Forest Functional Level: Windows Server 2016', 'Domain Functional Level: Windows Server 2016', and 'Additional Options: Global catalog: Yes, DNS Server: Yes, Create DNS Delegation: No'. At the bottom of the text box, there is a 'View script' button. Below the text box, there is a link to 'More about installation options'. At the bottom right, there are buttons for '< Previous', 'Next >', 'Install', and 'Cancel'. The target server is identified as 'DC01.tfsi.com'.

Active Directory Domain Services Configuration Wizard

TARGET SERVER
DC01.tfsi.com

Review Options

Review your selections:

Configure this server as the first Active Directory domain controller in a new forest.

The new domain name is "tfsi.com". This is also the name of the new forest.

The NetBIOS name of the domain: TFSI

Forest Functional Level: Windows Server 2016

Domain Functional Level: Windows Server 2016

Additional Options:

Global catalog: Yes

DNS Server: Yes

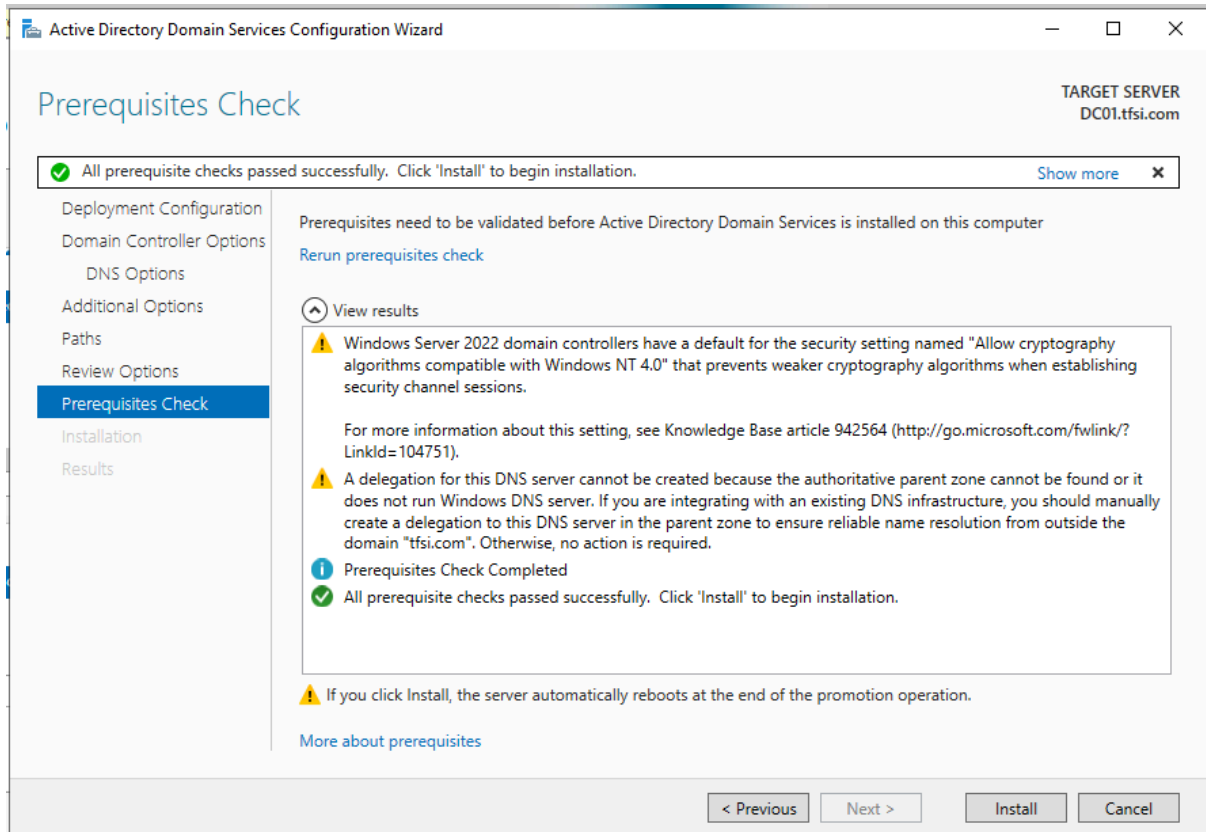
Create DNS Delegation: No

[View script](#)

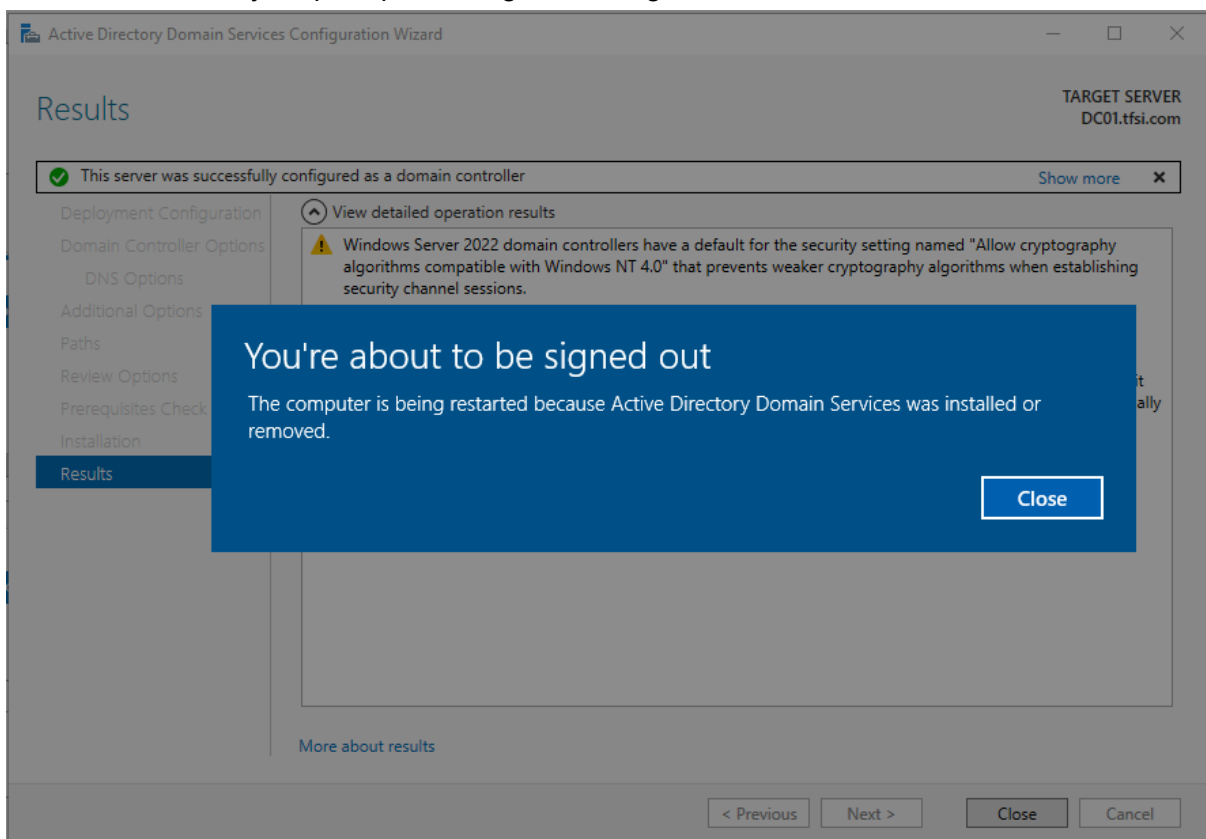
[More about installation options](#)

< Previous Next > Install Cancel

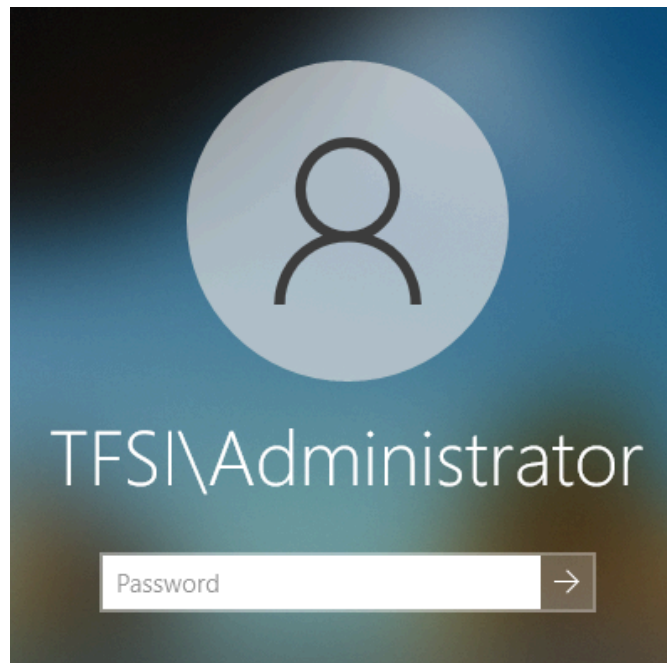
See if prerequisite checks are passed, Then click on Install



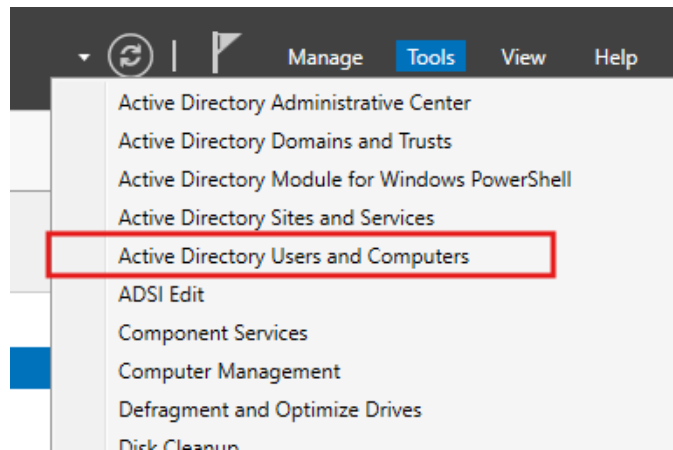
You will immediately be prompted to log off and login as AD DS server



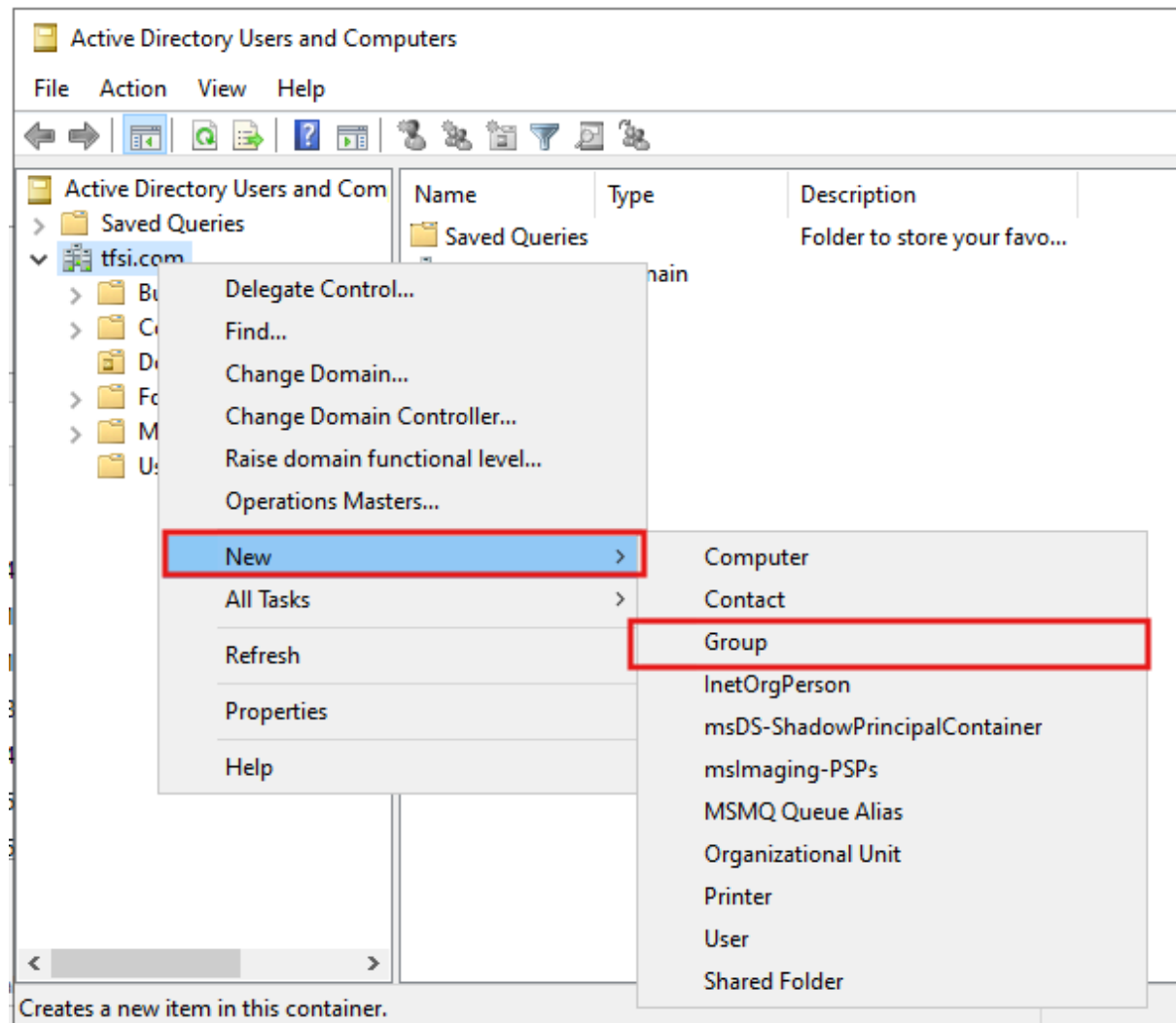
The Login username shows us as the NetBIOS/Username, showcasing that we are logging in the AD DS server.



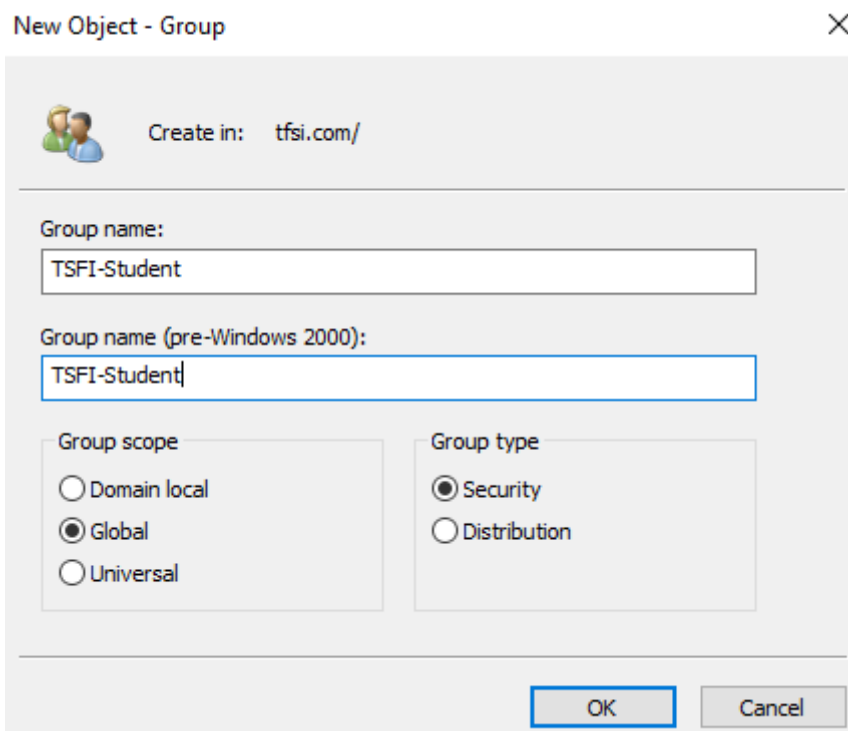
To create new Organizational Units like Groups and Users.



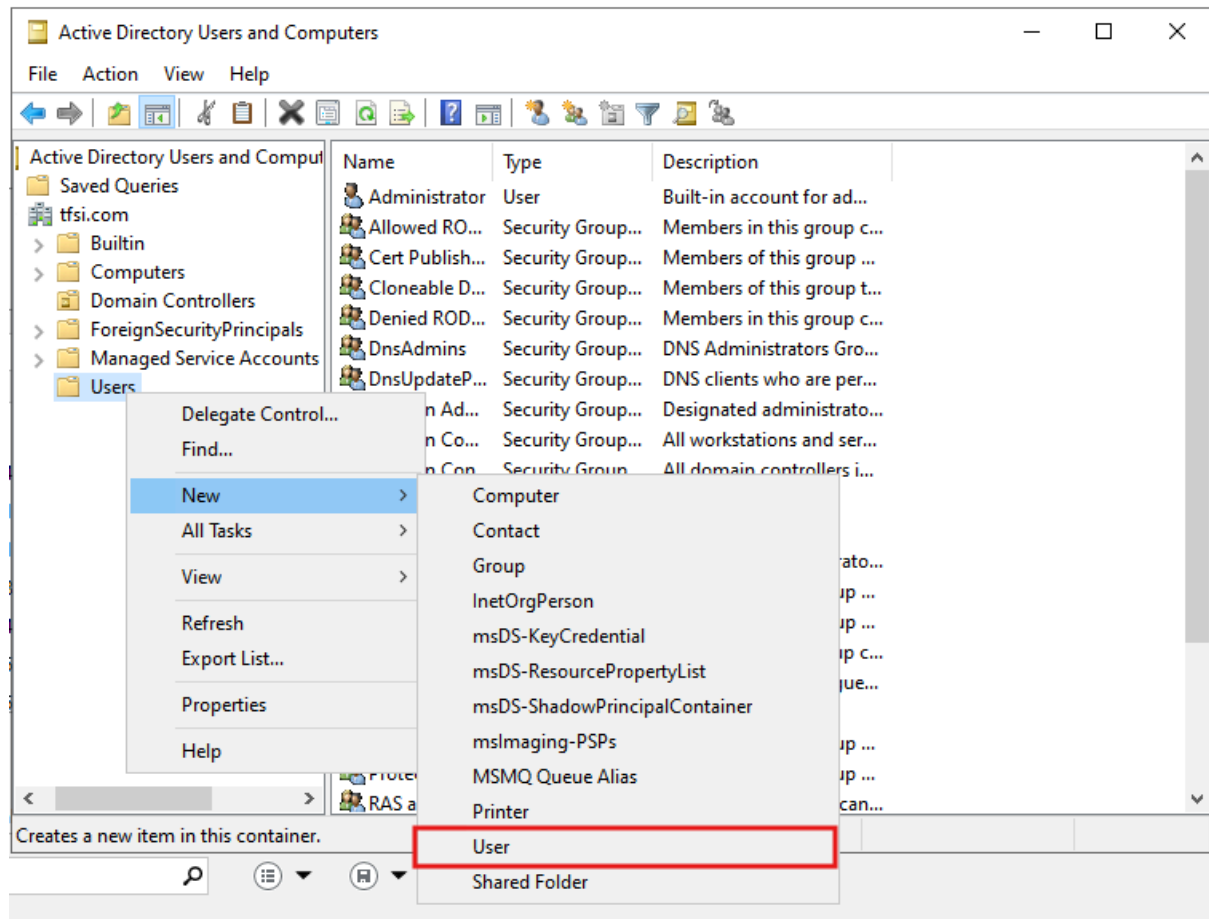
Click on your domain name, click on New > Group to create



Enter group details and click on OK.



In the users entry, Right Click > New > Users

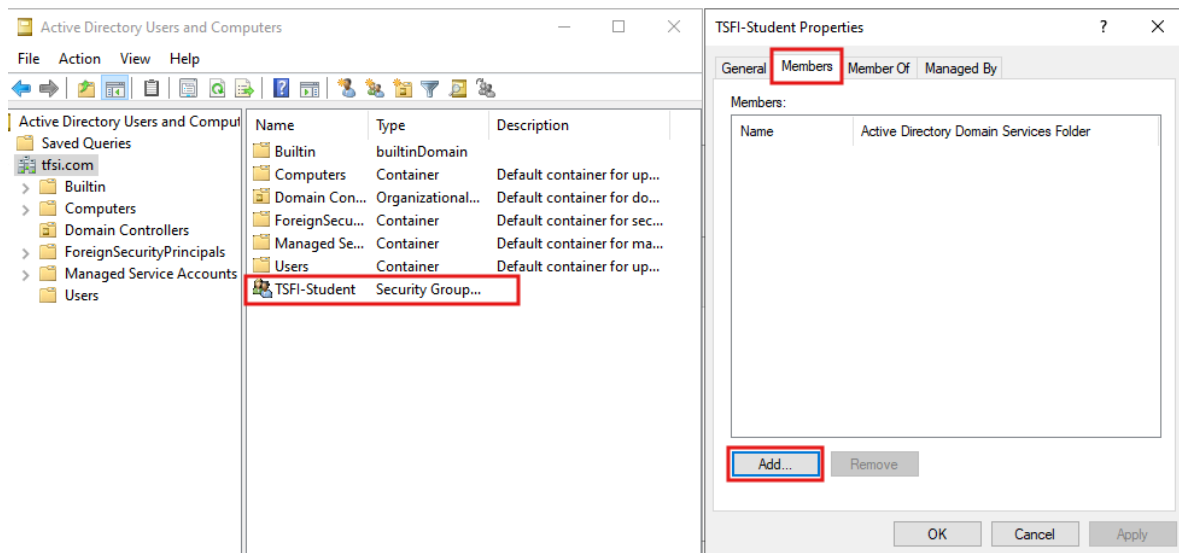


Fill in new user details.

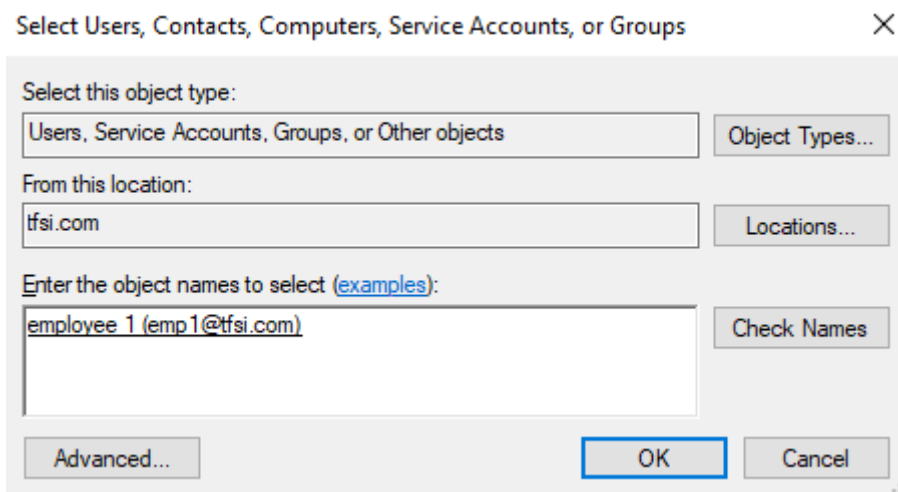
The first screenshot shows the 'New Object - User' wizard, Step 1: General. The 'Create in' field is set to 'tfsi.com/Users'. The 'First name' field contains 'employee', 'Last name' contains '1', and 'Full name' is 'employee 1'. The 'User logon name' is 'emp1' with a domain dropdown set to '@tfsi.com'. The 'User logon name (pre-Windows 2000)' is 'TFSI\emp1'. The 'Next >' button is highlighted.

The second screenshot shows the 'New Object - User' wizard, Step 2: Password. The 'Create in' field is 'tfsi.com/Users'. The 'Password' and 'Confirm password' fields are filled with masked characters. The 'User must change password at next logon' checkbox is checked. The 'Next >' button is highlighted.

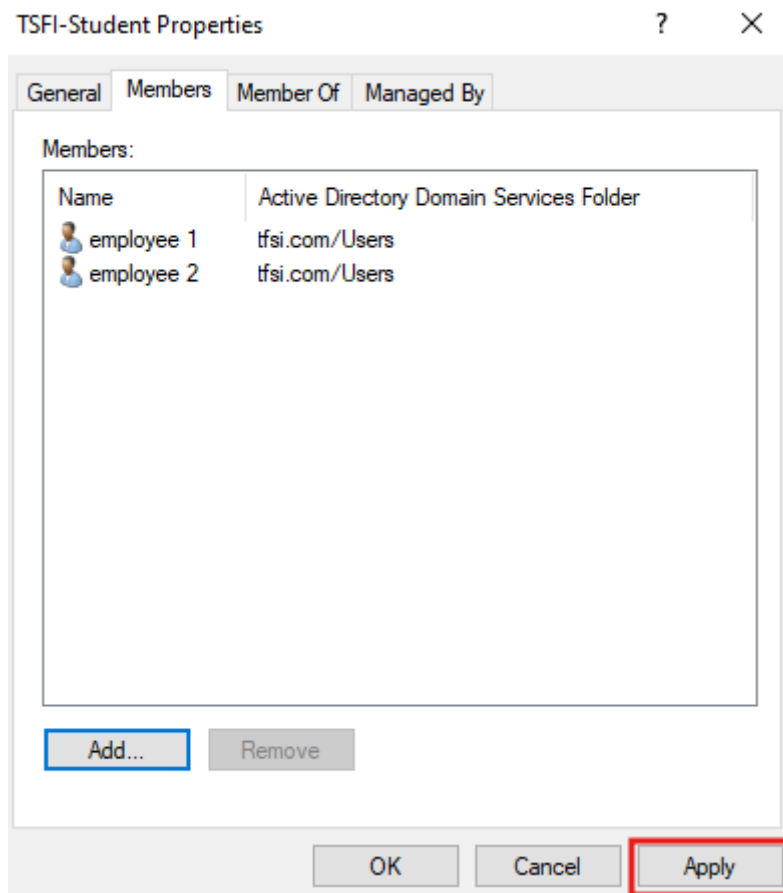
Now to add a new user to the group, select the group and view properties.
In the Member Tab



Add users objects and click on OK



Click on apply to add these members to the Group



Question 3)

- **Database folder:** Houses the core database files (such as **ntds.dit** in Active Directory), containing the actual directory data, objects, and attributes that power authentication and directory services.
C:\WINDOWS\NTDS
- **Log file folder:** Stores transactional logs (**edb.log** and checkpoint files) that record all changes made to the database prior to being written, ensuring data integrity and enabling recovery in the event of a system crash.
C:\WINDOWS\NTDS
- **SYSVOL folder:** A shared directory on domain controllers that replicates public files, such as Group Policy objects and logon scripts, across all domain controllers in the domain to maintain consistent policies and configurations.
C:\WINDOWS\SYSVOL

Question 4)

Feature	Workgroup (Peer-to-Peer)	Domain (Client-Server)
Centralization	Decentralized; each computer manages its own accounts, security, and policies.	Centralized; managed from a central server (Domain Controller) using services like Active Directory.
Scalability	Best suited for small networks (typically 10 or fewer computers).	Highly scalable, supporting thousands of users and devices across multiple locations.
User Accounts	Users must have a separate local account created on every single computer they want to access.	Users have a single network-wide account to log into any joined computer.
Security Management	Basic and managed locally per machine; changing a password requires updating it on each device separately.	Centralized and robust; administrators can enforce password policies, permissions, and group policies from one place.
Resource Sharing	Permissions must be configured individually for each shared folder or printer on every machine.	Centralized control over access to shared resources, files, and network printers.

Question 5)

DNS functions as both the navigational backbone and the locator service for the entire network. Rather than just translating human-readable computer names into IP addresses, Active Directory relies directly on DNS to help clients and domain controllers find each other using specialized service (SRV) records, such as locating authentication servers, global catalogs, and domain controllers across sites. DNS also is tightly integrated with Active Directory itself, enabling secure dynamic updates and replication between domain controllers so that changes synchronize efficiently across the forest.

Question 6)

The Directory Services Restore Mode (DSRM) password is a specialized local administrator credential used to boot a domain controller into an offline recovery state where Active Directory services are stopped. Administrators use it during disaster recovery scenarios, such as restoring a corrupted **ntds.dit** database, performing authoritative restores of deleted objects, or repairing directory components by interrupting the normal boot sequence, selecting DSRM, logging in with these dedicated credentials, and executing low-level maintenance utilities like **ntdsutil**.

Question 7)

Check the **DNS configuration** on the Windows 10 client first.

Because, A successful ping proves that basic IP-level network connectivity is working and the machine can reach **192.168.10.10**. However, joining a domain relies fundamentally on name resolution and specialized service (SRV) records rather than raw IP addresses. If the client's DNS server settings are misconfigured (e.g., pointing to an external DNS provider instead of the domain controller or internal DNS server), the client will be unable to resolve the **tfsi** domain name or find the necessary SRV records required to complete the join process.

Question 8)

The user account **Emp1** may be unable to log into the Windows 10 client due to a few common Active Directory authentication issues. If the client machine cannot properly communicate with the domain controller which is often caused by incorrect DNS server settings or a broken secure channel trust relationship between the workstation and the domain, the authentication requests will fail.